

MUSKAN YADAV

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SUMMARY

Data Analyst | AI Enthusiast | Integrated M.Tech (CSE – Computational & Data Science), VIT Bhopal (CGPA: 7.98/10). Skilled in Python, SQL, Power BI, Machine Learning, Data Analytics, Data Visualization, Excel, and AI. Built 3+ end-to-end AI & software projects, transforming data into actionable insights and developing scalable, data-driven solutions.

EDUCATION

Vellore Institute of Technology, Bhopal
Integrated M.Tech (B.Tech + M.Tech) in CSE

2022 – 2027
CGPA: 7.98/10

EXPERIENCE

Data Science Intern

Dec 2024 – Feb 2025

CodeSoft

- Developed machine learning pipelines using Scikit-learn; solved 5+ real-world business problems through iterative model tuning.
- Performed data analysis and manipulation using Pandas and NumPy across multiple structured datasets to surface key trends.
- Executed ETL workflows — preprocessing and feature engineering — across 4+ datasets; improved data accuracy by 15%.
- Validated data quality and consistency, ensuring reliable inputs for downstream analysis and modeling.
- Created statistical predictive models that informed data-driven business decision-making.
- Crafted Matplotlib and Excel visualizations to communicate insights; cut manual reporting time by 20%.

TECHNICAL SKILLS

Programming Languages: Python, SQL

Data Science & ML: Power BI, Pandas, NumPy, Scikit-learn, XGBoost, Random Forest, SARIMA, Prophet, PyTorch, LSTM, Time Series Forecasting, EDA, Predictive Analytics

AI & Generative AI: LLMs, RAG, AI Agents, Prompt Engineering, Groq API, Llama 3

Databases: MySQL, PostgreSQL, MongoDB

Developer Tools: Git, GitHub, Docker, MLflow, VS Code, Jupyter Notebook

Core CS: DSA, OOP, DBMS, Operating Systems, Computer Networks

PROJECTS

Retail Demand Forecasting & Inventory Optimization | Python, XGBoost, Prophet, PyTorch, FastAPI, Streamlit, Docker | URL

- Engineered a forecasting engine on 3.5 years of retail data across 25 store-department segments; benchmarked 4 architectures (SARIMA, Prophet, LSTM, XGBoost).
- Achieved 4.74% MAPE using global XGBoost with lagged/rolling features, outperforming Prophet (4.93%) and LSTM (10.37%).
- Optimized EOQ, Safety Stock, and ROP parameters; achieved 15–20% cost savings in a 26-week simulation.
- Deployed 2 microservices (FastAPI + Streamlit) via Docker Compose; tracked experiments with MLflow.

Customer Shopping Behaviour Analysis | Python, SQL, Power BI, NumPy, Matplotlib | URL

- Analyzed 3,900+ customer transaction records using Python and Pandas to clean data and identify purchasing patterns.
- Developed SQL analytical queries using filtering, grouping, and aggregations to evaluate revenue, sales KPIs, and customer behaviour across multiple dimensions.
- Built an interactive Power BI dashboard with 10+ KPIs and visualizations to analyze revenue, customer segments, and subscription trends.

MedAgent — AI Medical Assistant | Python, Groq API, Llama 3, RAG, Streamlit | URL

- Architected an agentic AI medical assistant using RAG and the Llama 3 LLM to answer health queries from medical documents.
- Optimized inference using Groq's LPU-based architecture; reduced response time to under 2 seconds per query, a 4x speedup.
- Enabled real-time RAG-based retrieval across 100+ medical topics, improving query coverage and accuracy.
- Maintained multi-turn conversational context using session memory across 10+ exchanges.

CERTIFICATES & ACHIEVEMENTS

- Artificial Intelligence Fundamentals – IBM SkillsBuild (2024)
- Applied Machine Learning in Python – Coursera (2023)
- HackerRank 5-Star in Python – Solved 100+ algorithmic problems
- Volunteered with the NSS Club, leading 5+ awareness drives and coordinating 3+ campus events.
- Kanchan Education and Welfare Society: Supported welfare programs reaching 50+ beneficiaries.